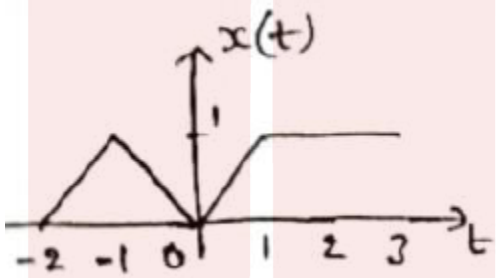
Disclaimer: These questions are picked by the TIE review team teachers/mentors by referring to more than 15 previous years question papersand Internal question papers from more than 10 colleges. The sole purpose of this is to give a thorough idea about the type of Questions in the final assessment paper(sem-end exams).We kindly request the students to practice problems/theory of these types(min 3-4 questions per question type) and other types as well for the best results.

SS-IMP and SIMP Topics-18Scheme

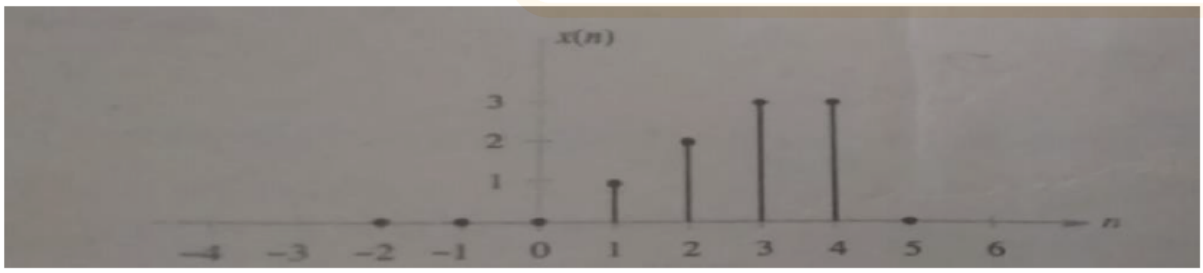
By the ECE TIE review team

Module-1 to 5 SIMP

1. Define signals. explain the classification of signals, discuss the operations on signals (i)Differentiation and Integration
2. Determine whether the given continuous-time signal is periodic or not if periodic find the fundamental period x(t)=cos100πt+ sin50πt
3. Determine and sketch the even and odd components of the signal



4. A continuous time signal x(t) is shown in figure sketch and label each of the following signals (i) x(t-2) (ii) x(2t) (iii)x(t/2)



5. Prove that the product of two even signals is even and product of an even and odd signal is odd, also Evaluate the following Integrals

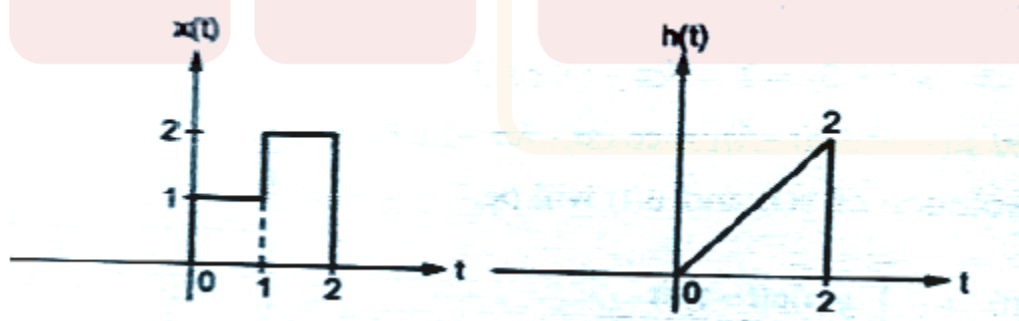
$$i)\ \int_{-1}^{1}\left(\frac{t^2}{2}+5\right)\delta(t)\ dt \qquad ii)\ \int_{-\infty}^{\infty}(\sin \pi t + t\cos^2 \pi t)\,\delta(t-1)\,dt$$

$$iii)\ \int_{-\infty}^{\infty} e^{-t}\,\delta(2t-1)\,dt \qquad iv)\ \int_{-3}^{-1} e^{-t}\,\delta(t)\ dt$$

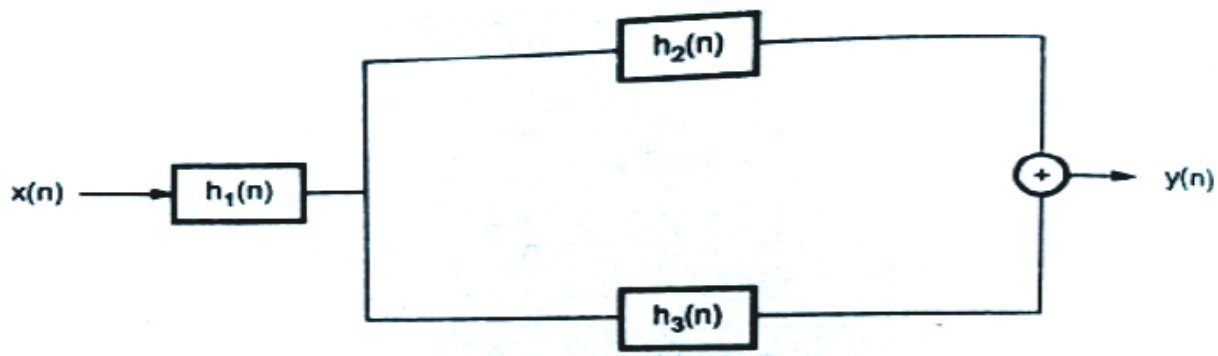
6. Discuss the properties of systems, Find and sketch the first derivatives of the signal x(t)=u(t)-u(t-a), a>0
7. For a system described by T{x(n)} = ax(n)+b, check for the following properties (i) Causality (ii) Linearity
8. Determine which of the following system are invertible
   a. Y(n)=2x(n) b. Y(n)=x(n) -x(n-1)
9. Perform the convolution of two finite sequences using the graphical method only X(n)={-1, 1, 0, 1,-1} h(n)={1, 2, 3}
10. Convolute the following two continuous time signals

$$x1(t)=e^{-2t}u(t) \qquad x2(t)=u(t+2)$$

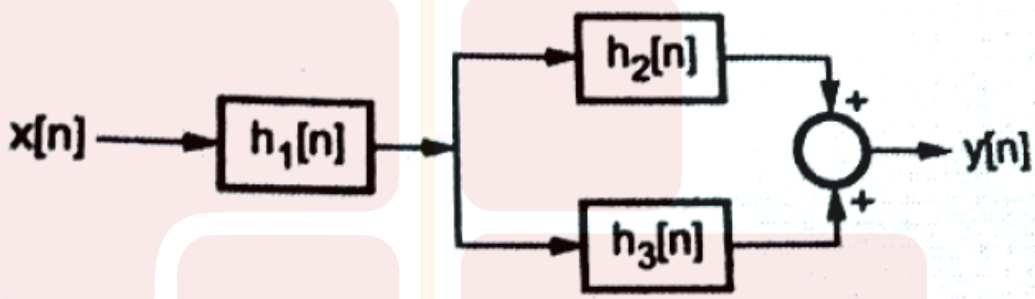
11. Perform the convolution of the two signals shown in Figure



12. Prove the commutative and associative properties of the convolution sum
13. Find the overall impulse response of a cascade of two systems having identical impulse responses h(t)=2{u(t)-u(t-1)}
14. The LTI systems are connected as shown in figure 3(b) If h1(n)=u(n-2), h2(n)=nu(n) and h3(n)= δ(n-2). Find the overall response



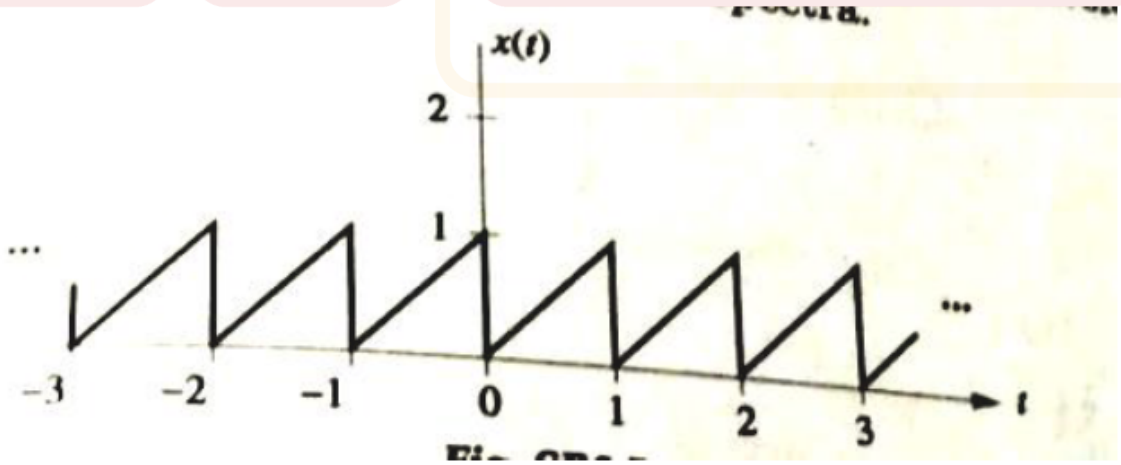
15. Compute y(n) as ahown in figure 4(a), for the given h1(n)=(1/2)n u(n); h2(n)=u(n-2); h3(n)=δ(n)+ δ(n-1).



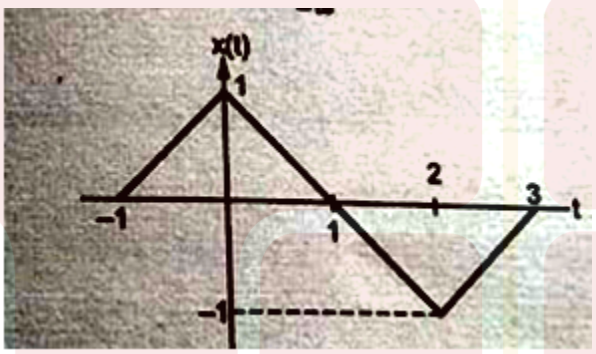
16. Find the step response of an LTI system, whose impulse response is given by

$$h(t)=t^2u(t)$$

17. State and prove the following properties of CTFS (i) Frequency shift (ii)Scaling, Find the complex Fourier Coefficient for x(t) given below X(t)=cos(2π/3 t) +2 cos(5π/3 t)
18. State and prove multiplication theorem
19. Find the Complex Fourier Coefficient for the periodic waveform x(t) shown in figure Also draw the amplitude and phase spectra



20. State and prove the following properties of Z transform (i). Differentiation in Z domain, (ii). Convolution in time domain, (iii). Time reversal.
21. Find the ZT of x(n)={ 1,2, 3, 4,5,0,7}.
22. Find the ZT of $x(n)= a^{|n|}$, 0<a<1 and find the Region of Convergence
23. Discuss the properties of Region of Convergence (ROC), Find the Z Transform, ROC and Poles and zeros of the signal $x(n) = \left(\frac{1}{3}\right)^n \sin\left(\frac{\pi n}{4}\right) u(n)$.
24. Using the properties of DTFT evaluate $x[n]=\alpha^n \sin(\Omega_0 n)u(n)$, for $\alpha<1$
25. State and prove frequency differentiation property and parsevals theorem for DTFT.
26. For the signal x(t) shown, evaluate $\int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$ without explicitly computing X(w)



27. State and prove Integration and scaling property of CTFT.
28. Using the properties of the Z Transform, find the Z transform of these signals $x(n)=n^2(1/2)^n u(n-3)$.
29. A system has impulse response $h(n)=(1/2)^n u(n)$.Determine the input to the system if the output is given by $y(n)=1/3u(n)+2/3\ (-1/2)^n u(n)$.
30. Find the Inverse Z transform of the following

$$X(Z)= \frac{Z^3+ Z^2+ 3/2\ Z +1/2}{Z^3+ 3/2\ Z^2+1/2\ Z} \quad \text{for } Z<1/2.$$